

Symmetric Matrices

Geometric Algorithms

Keywords

linear models

design matrices

general linear regression

symmetric matrices

the spectral theorem

orthogonal diagonalizability

quadratic forms

definiteness

constrained optimization

Recall: Symmetric Matrices

Def. A square matrix A is **symmetric** if $A^T = A$

$$\begin{bmatrix} 1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Orthogonal Eigenvectors

Thm. For a symmetric matrix A , if u and v are eigenvectors for *distinct* eigenvalues, then u and v are orthogonal

$$Au = \lambda_1 u$$

$$Av = \lambda_2 v$$

$$\lambda_1 \neq \lambda_2$$

$$\langle Au, v \rangle =$$

$$\langle \lambda_1 u, v \rangle =$$

$$\langle Au, v \rangle =$$

$$(Au)^T v =$$

$$u^T A^T v =$$

$$u^T A v =$$

$$u^T (\lambda_2 v) =$$

$$\lambda_2 u^T v = \lambda_2 \langle u, v \rangle$$

$$\lambda_1 \langle u, v \rangle = \lambda_2 \langle u, v \rangle$$
$$\Rightarrow (\lambda_1 - \lambda_2) \langle u, v \rangle = 0$$
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Recall: Diagonalizable Matrices

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There is an invertible matrix P and diagonal matrix D such that $A = PDP^{-1}$

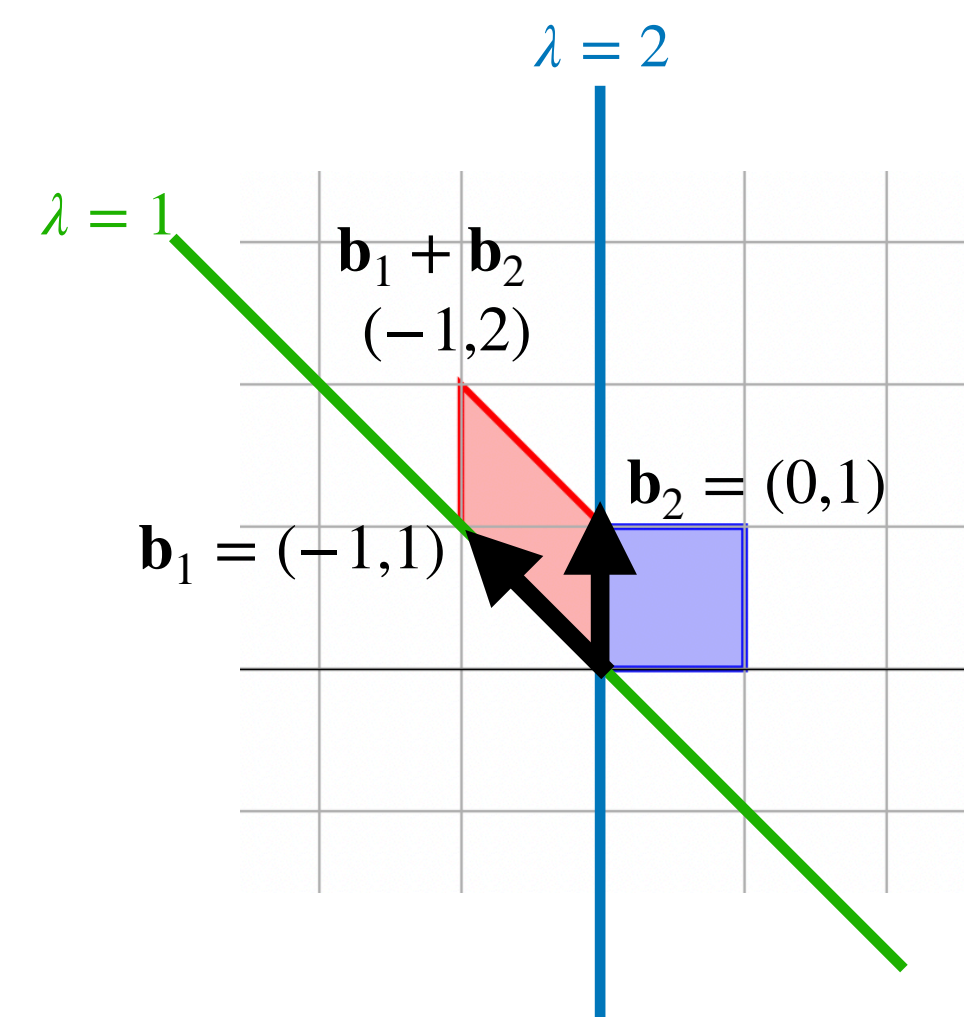
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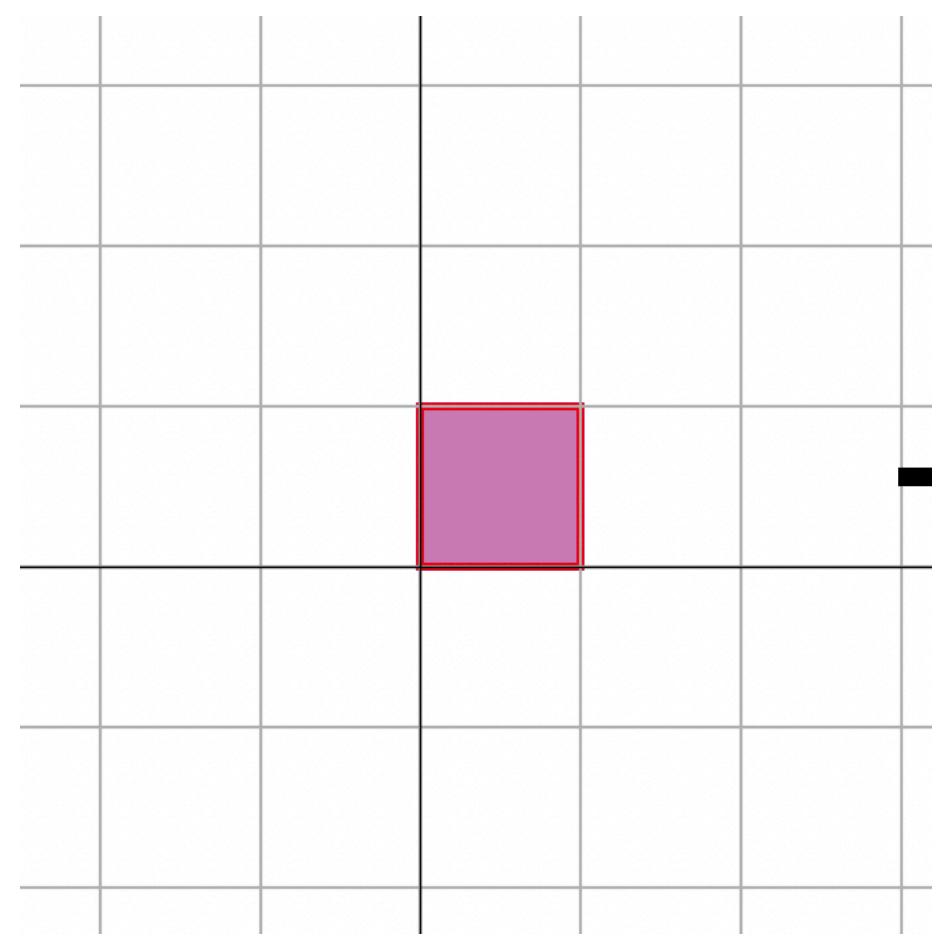
Diagonalizable matrices are the same as scaling matrices up to a change of basis

Recall: The Picture

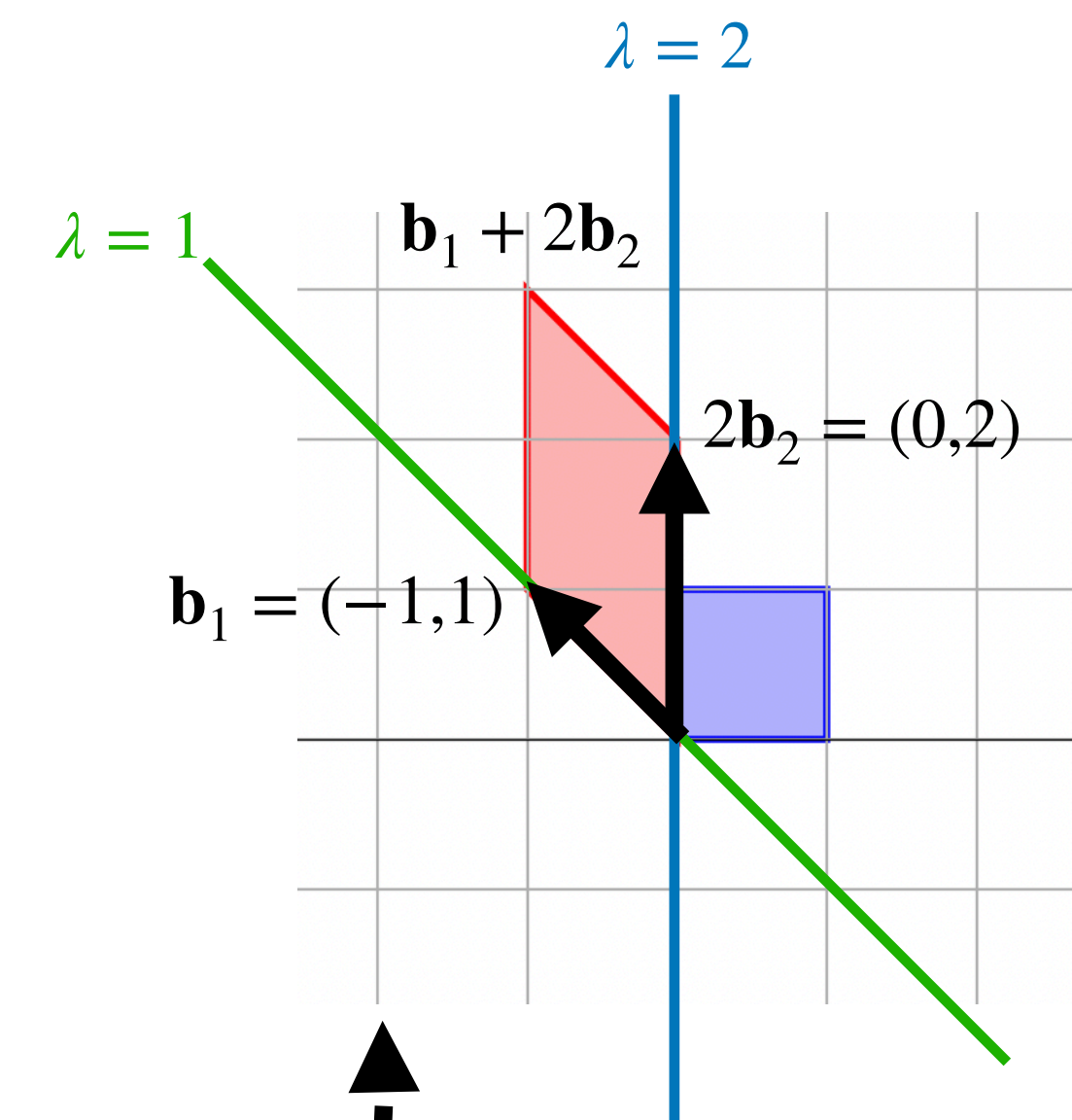
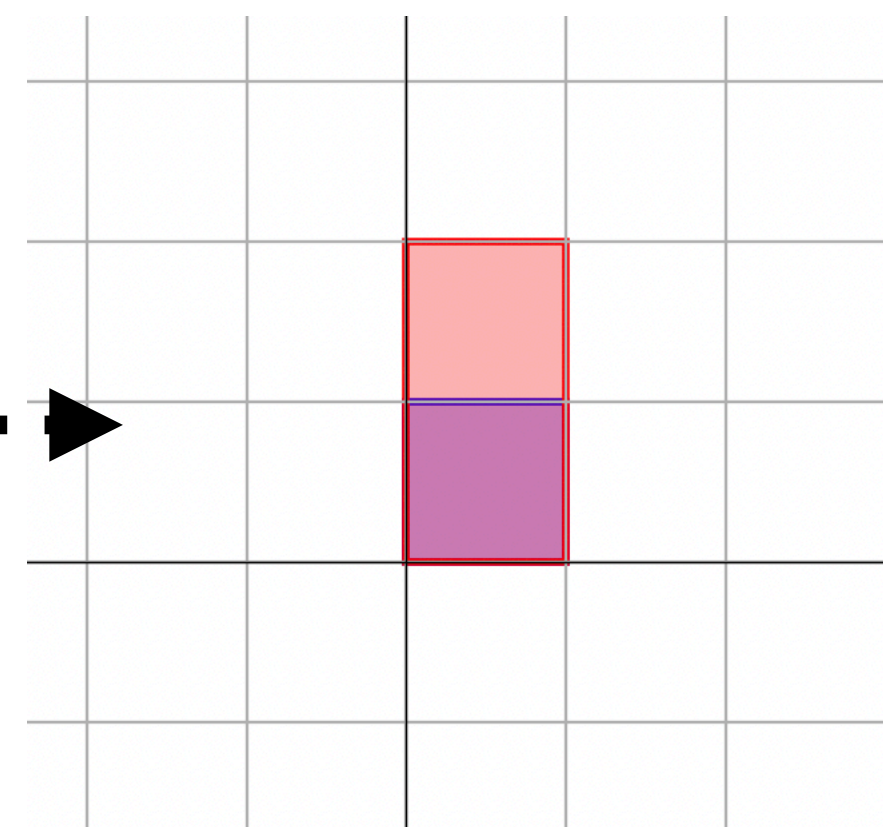


$$A = PDP^{-1}$$
$$\begin{bmatrix} 2 & 0 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & 1 \end{bmatrix}^{-1}$$

P^{-1}



D



P

Recall: The Diagonalization Theorem

$$A = PDP^{-1}$$

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The idea:

Recall: The Diagonalization Theorem

$$A = \overset{\text{eigenbasis}}{P} D P^{-1}$$

Thm. A is diagonalizable $\Leftrightarrow A$ has an eigenbasis

The idea:

» The columns of P form an **eigenbasis** for A

Recall: The Diagonalization Theorem

$$A = \overset{\text{eigenbasis}}{P} \overset{\text{eigenvalues}}{D} P^{-1}$$

Thm. A is diagonalizable $\Leftrightarrow A$ has an eigenbasis

The idea:

» The columns of P form an **eigenbasis** for A

» The diagonal of D are the eigenvalues for each column of P

Recall: The Diagonalization Theorem

$$A = \overset{\text{eigenbasis}}{P} \overset{\text{eigenvalues}}{D} P^{-1}$$

Thm. A is diagonalizable $\Leftrightarrow A$ has an eigenbasis

The idea:

- » The columns of P form an **eigenbasis** for A
- » The diagonal of D are the eigenvalues for each column of P
- » The matrix P^{-1} is a change of basis to this eigenbasis of A

The Spectral Theorem

Thm. If A is symmetric then it has an *orthonormal* eigenbasis

Symmetric matrices are *diagonalizable*, and we can choose P to be *orthonormal*

Recall: Orthonormal Matrices

Def. A matrix is **orthonormal** if its columns form an orthonormal set

This is sometimes called an **orthogonal** matrix

Recall: Orthonormal Matrices

Def. A matrix is **orthonormal** if its columns form an orthonormal set

This is sometimes called an **orthogonal** matrix

And this is incredibly confusing, but we'll try to be consistent and clear

Recall: Inverses of Orthogonal Matrices

Thm. If a matrix $U \in \mathbb{R}^{n \times n}$ is orthogonal then it is invertible and

$$U^{-1} = U^T$$

Orthogonal Diagonalizability

Def. A matrix A is **orthogonally diagonalizable** if there is a diagonal matrix D and matrix P such that

$$A = PDP^{-1} = PDP^T$$

P must be an *orthonormal matrix*

**Symmetric matrices are
orthogonally diagonalizable**

Orthogonal Diagonalizability and Symmetry

Thm. $A \in \mathbb{R}^{n \times n}$ is orthogonally diagonalizable if and only if A is symmetric

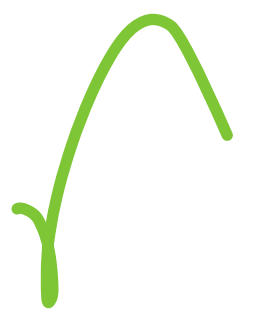
You should know how to...

- » Determine by looking at a matrix if it is orthogonally diagonalizable
- » Determine an orthonormal eigenbasis of a (small) symmetric matrix by hand
- » Find an orthogonal diagonalization of a (small) matrix by hand

Quadratic Forms

Quadratic Forms

Def. A quadratic form over the variables $x_1, \dots, x_n$ is an multivariate polynomial $Q(x_1, \dots, x_n)$ in which every term has degree two

$$x_1^2 + 3x_3x_1 + 2x_2^2$$


$$x_1^3 \quad \text{X}$$

$$x_1x_2x_3 \quad \text{X}$$

$$x_1 \quad \text{X}$$

Quadratic Forms and Symmetric Matrices

Fact. Quadratic forms can be represented as

$$Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$$

where A is *symmetric*

Example: Computing the Quadratic Form for a Matrix

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix}$$

This means, given a symmetric matrix A , we can compute its corresponding quadratic form

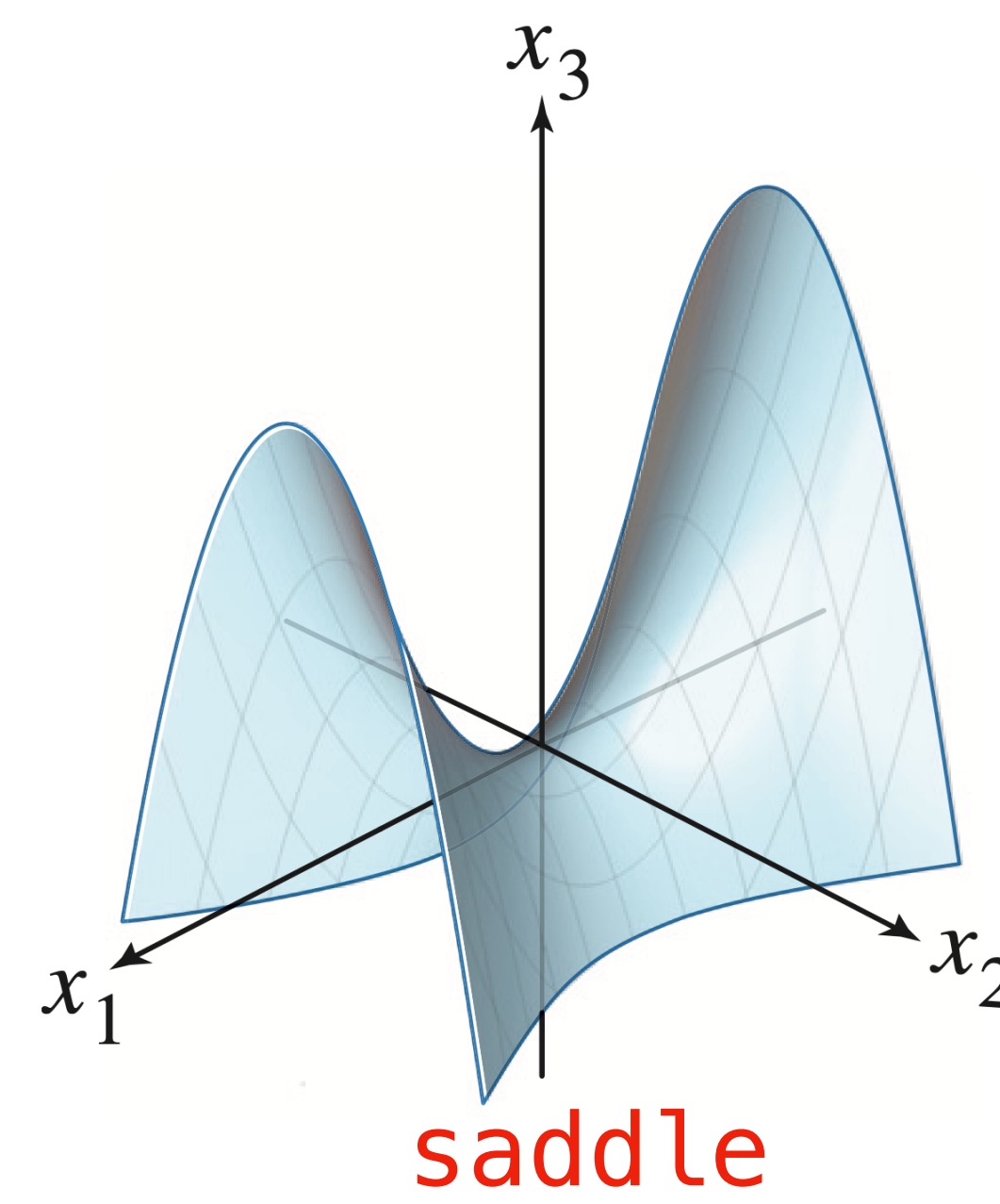
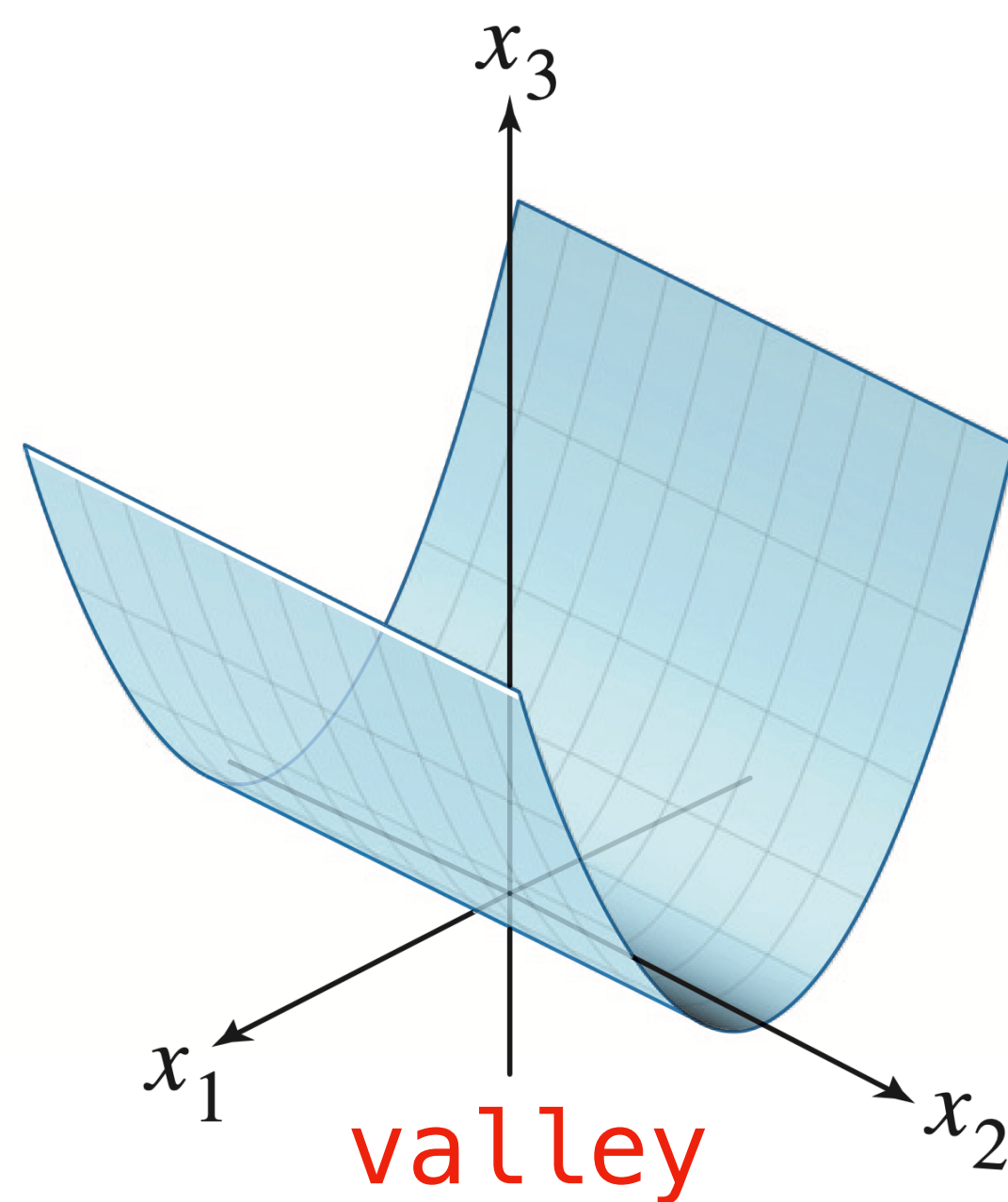
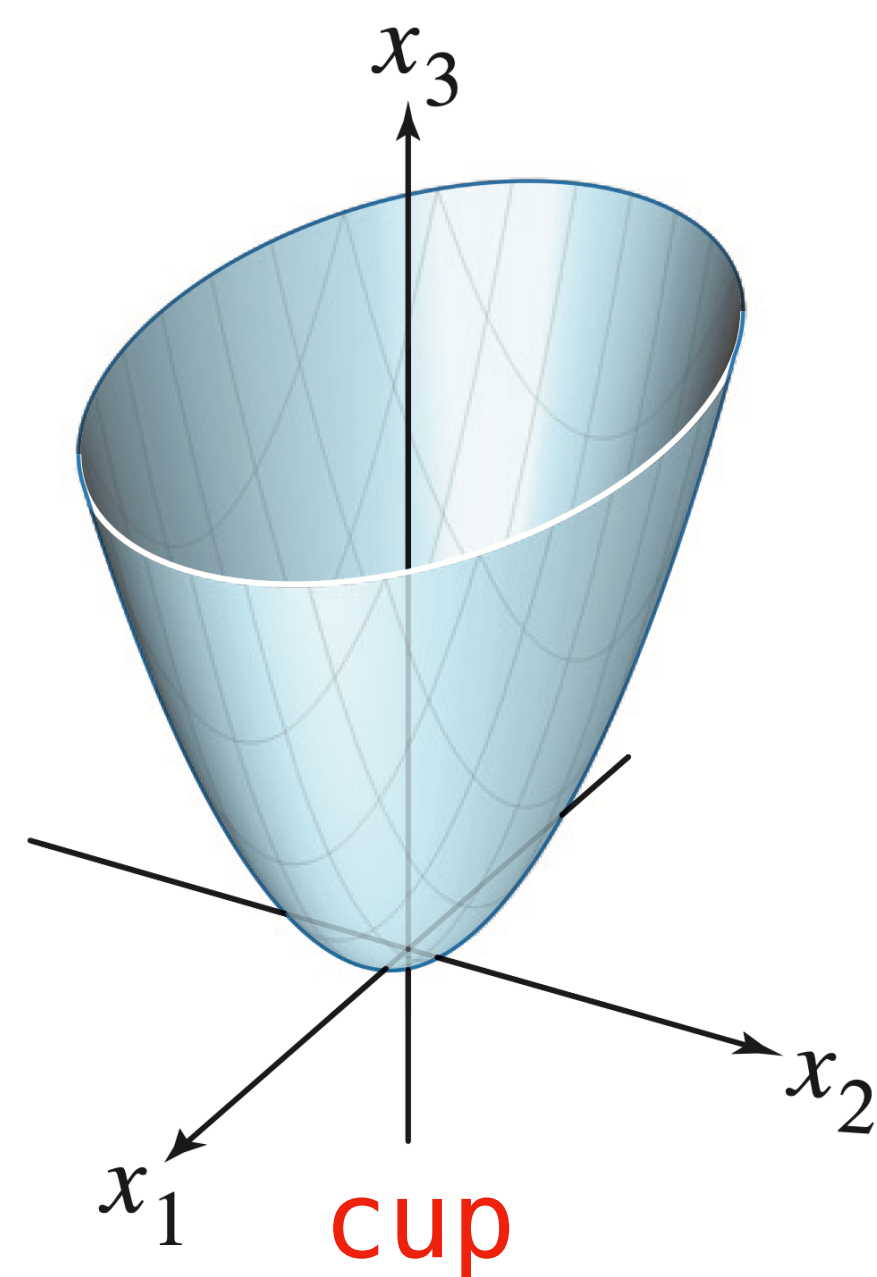
$$Q(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^T A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

In General

$$\mathbf{x}^T \mathbf{A} \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j = \sum_{i=1}^n A_{ii} x_i^2 + \sum_{i \neq j} (A_{ij} + A_{ji}) x_i x_j$$

$$\begin{aligned} \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a & b \\ b & d \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} x_1 & x_2 \end{bmatrix} * \begin{bmatrix} ax_1 + bx_2 \\ bx_1 + dx_2 \end{bmatrix} \\ &= x_1(ax_1 + bx_2) + x_2(bx_1 + dx_2) \\ &= ax_1^2 + bx_1x_2 + bx_2x_1 + dx_2^2 \\ &= ax_1^2 + 2bx_1x_2 + dx_2^2 \end{aligned}$$

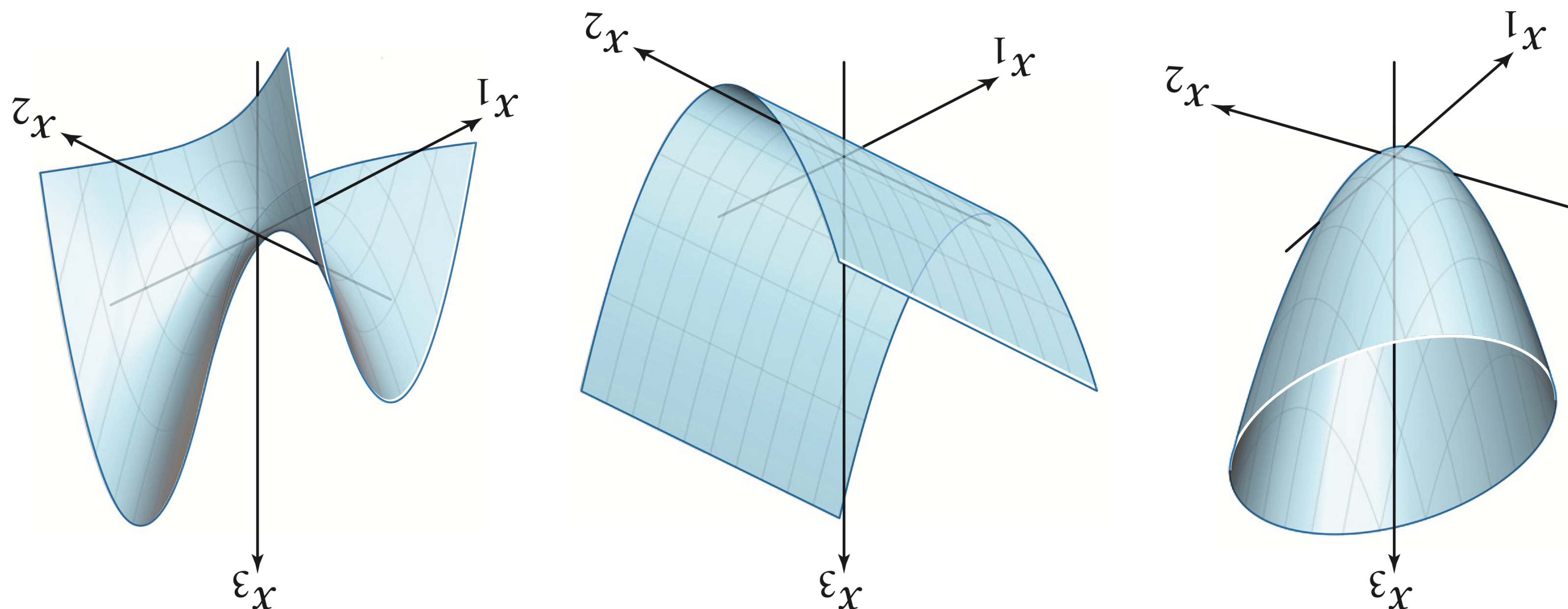
Shapes of of Quadratic Forms



There are three possible shapes (six if you include the negations)

Can we determine what shape it will be mathematically?

Shapes of of Quadratic Forms



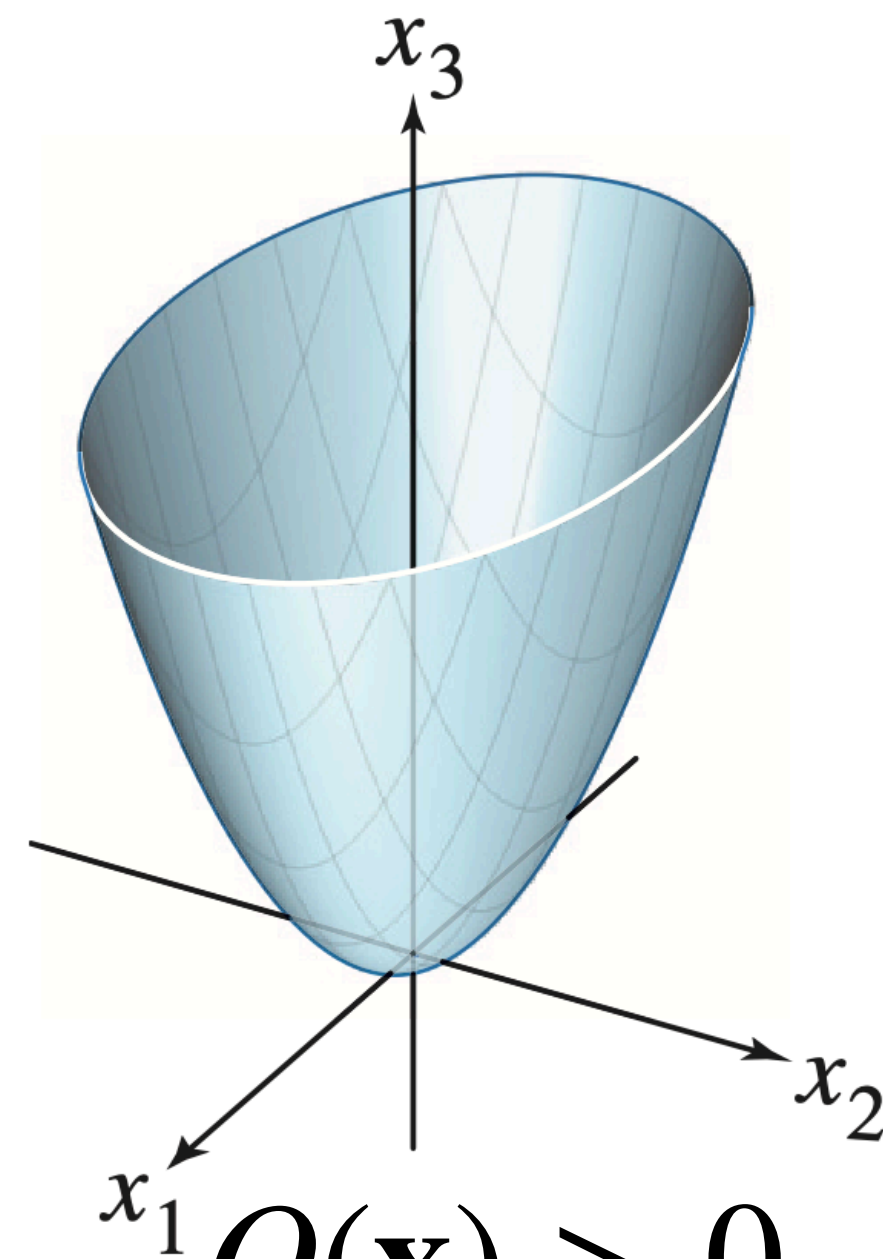
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Definiteness

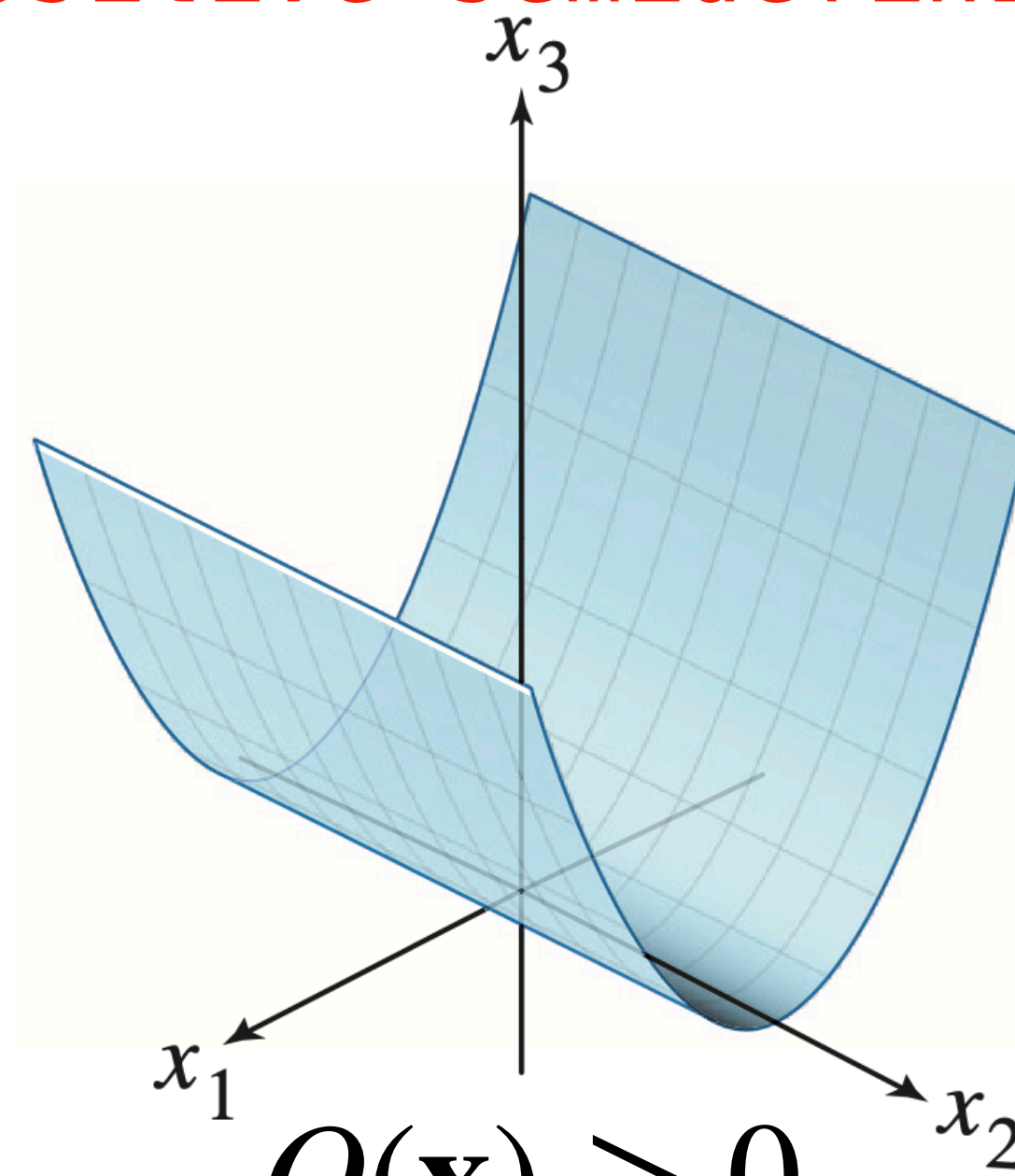
positive semidefinite

negative definite

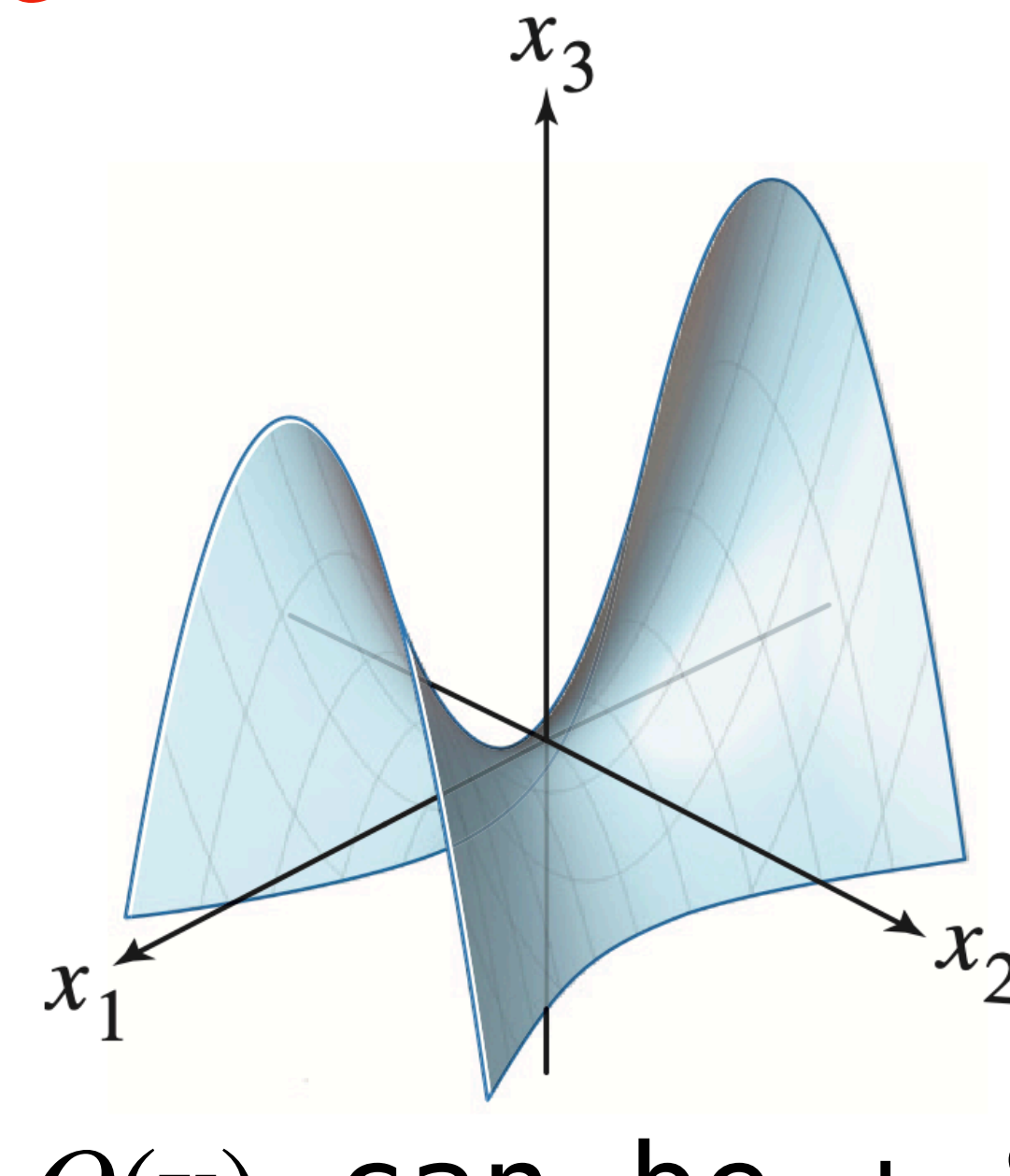


$$Q(\mathbf{x}) > 0$$

positive definite

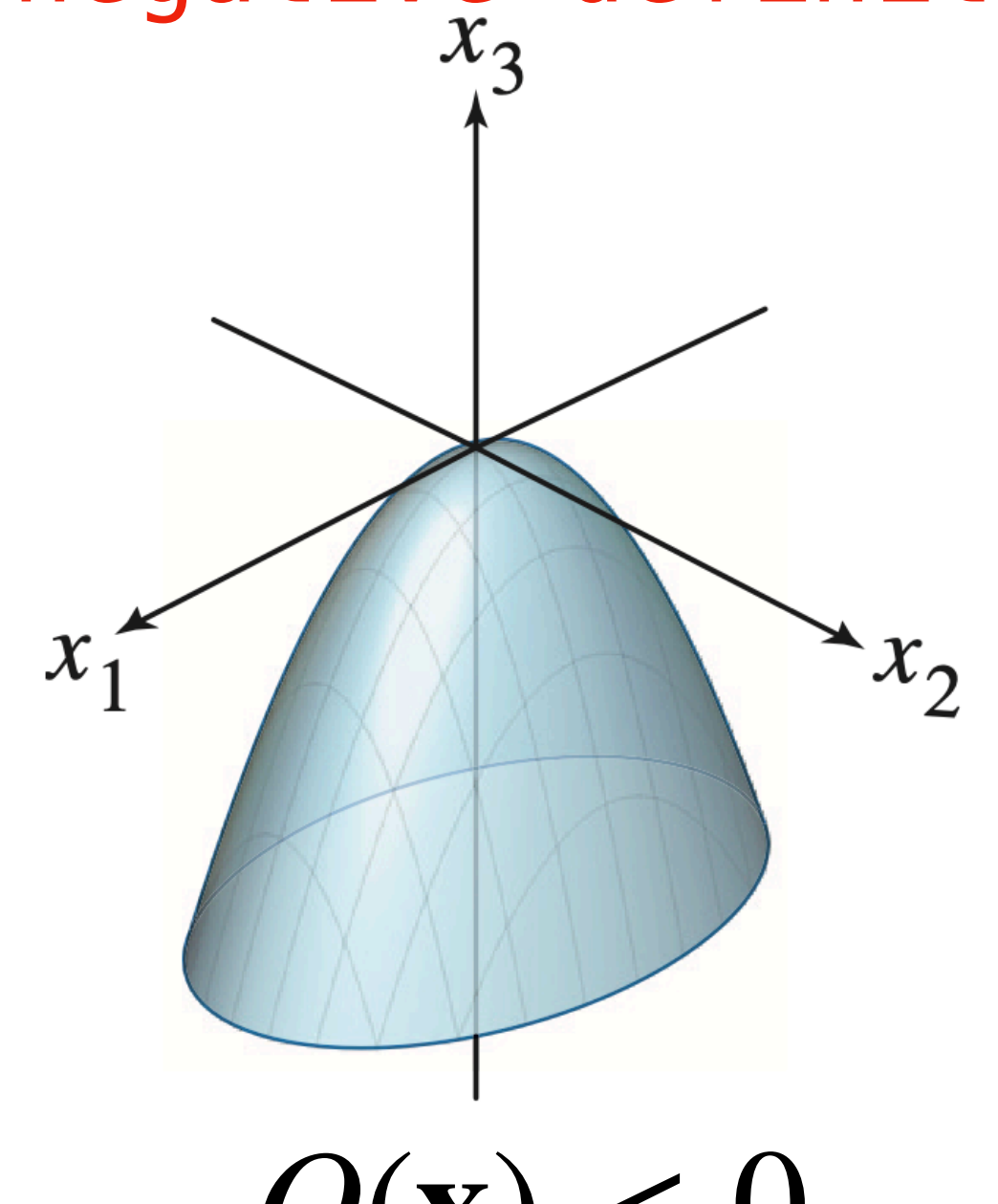


$$Q(\mathbf{x}) \geq 0$$



$Q(\mathbf{x})$ can be + & - $Q(\mathbf{x}) < 0$

indefinite



For $\mathbf{x} \neq \mathbf{0}$ the graphs satisfy the given properties

Definiteness and Eigenvectors

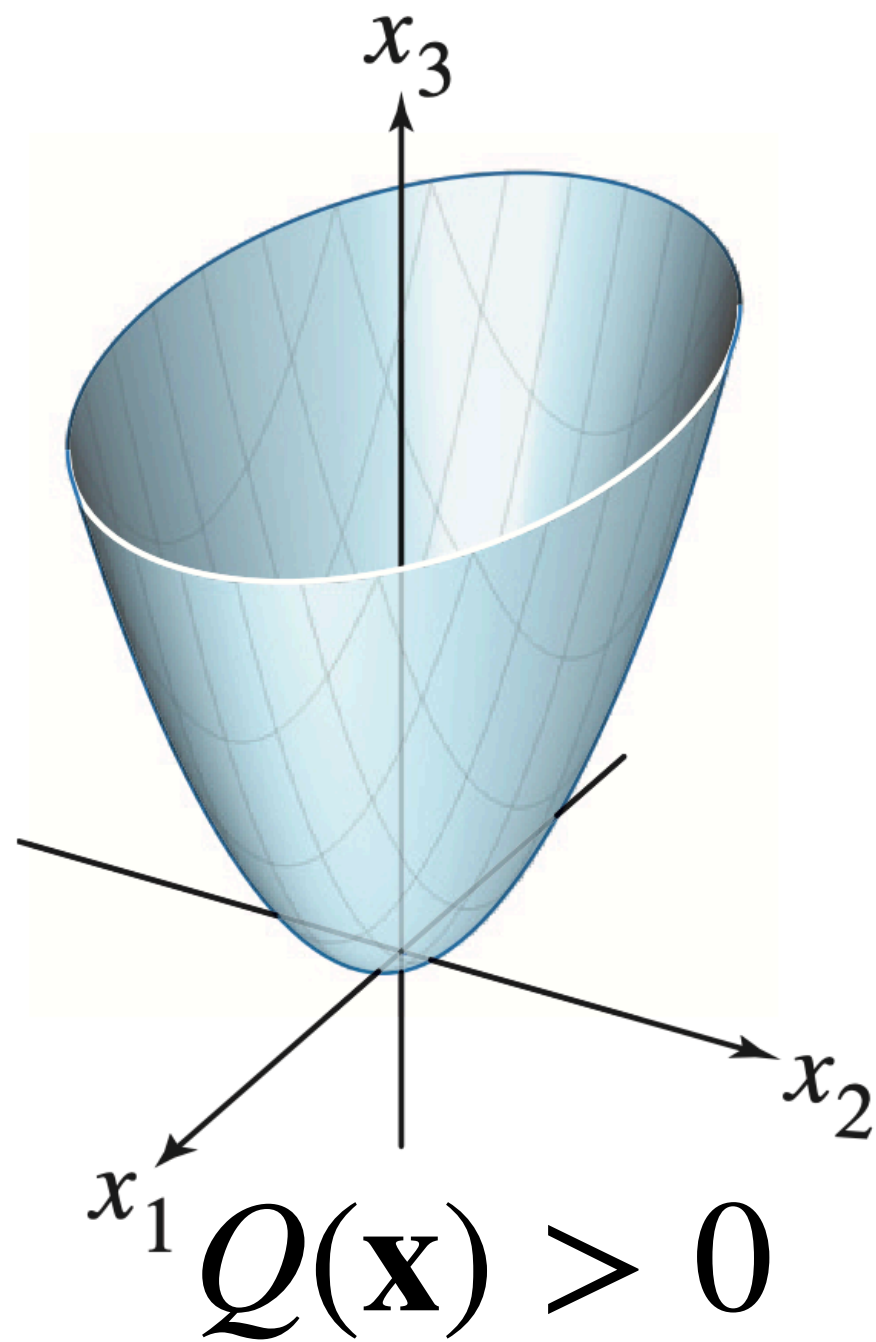
Thm. For symmetric A the quad form $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ is

- » **positive definite** $\equiv$ all *positive* eigenvalues
- » **positive semidefinite** $\equiv$ all *nonnegative* eigenvalues
- » **indefinite** $\equiv$ positive and negative eigenvalues
- » **negative definite** $\equiv$ all *negative* eigenvalues

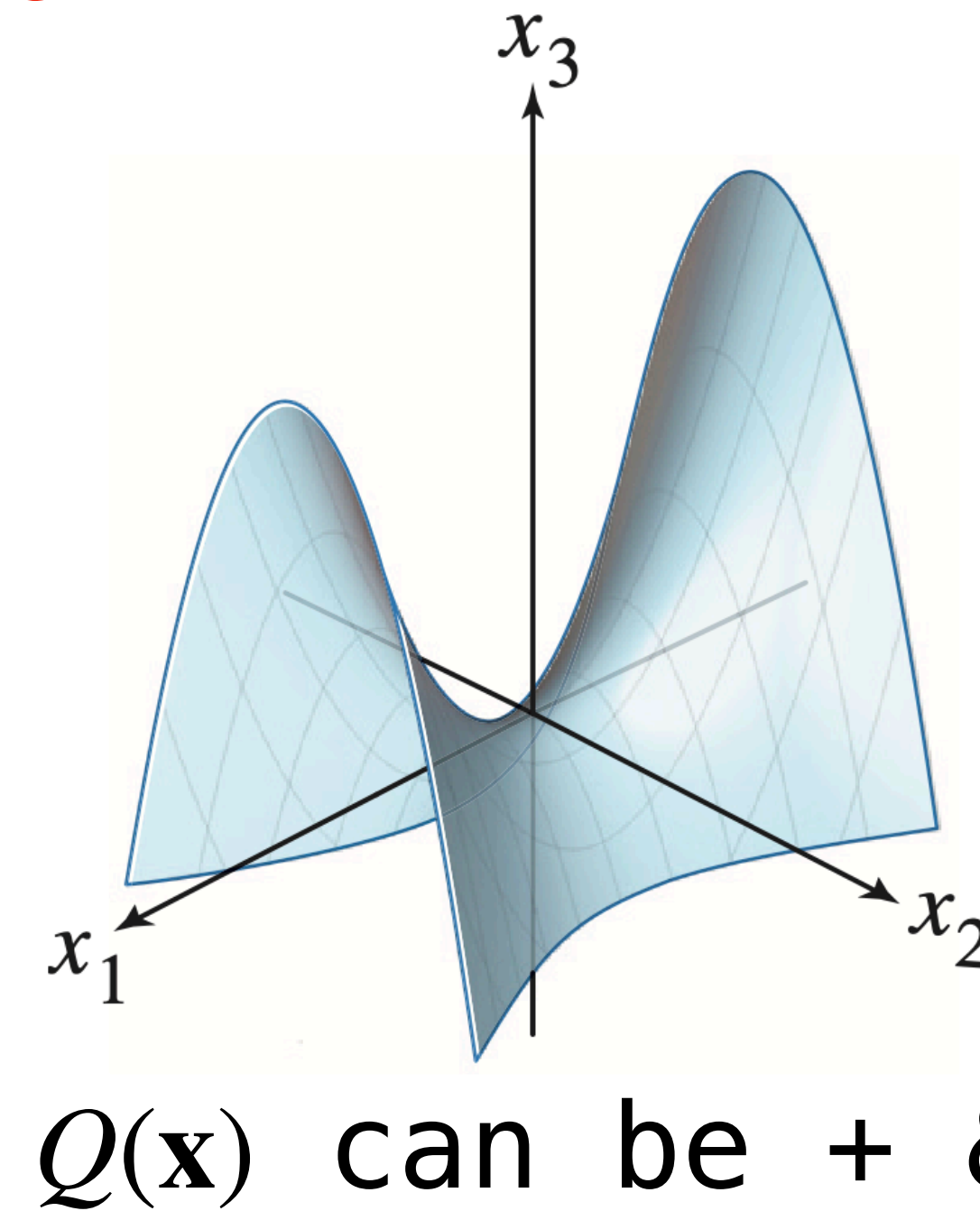
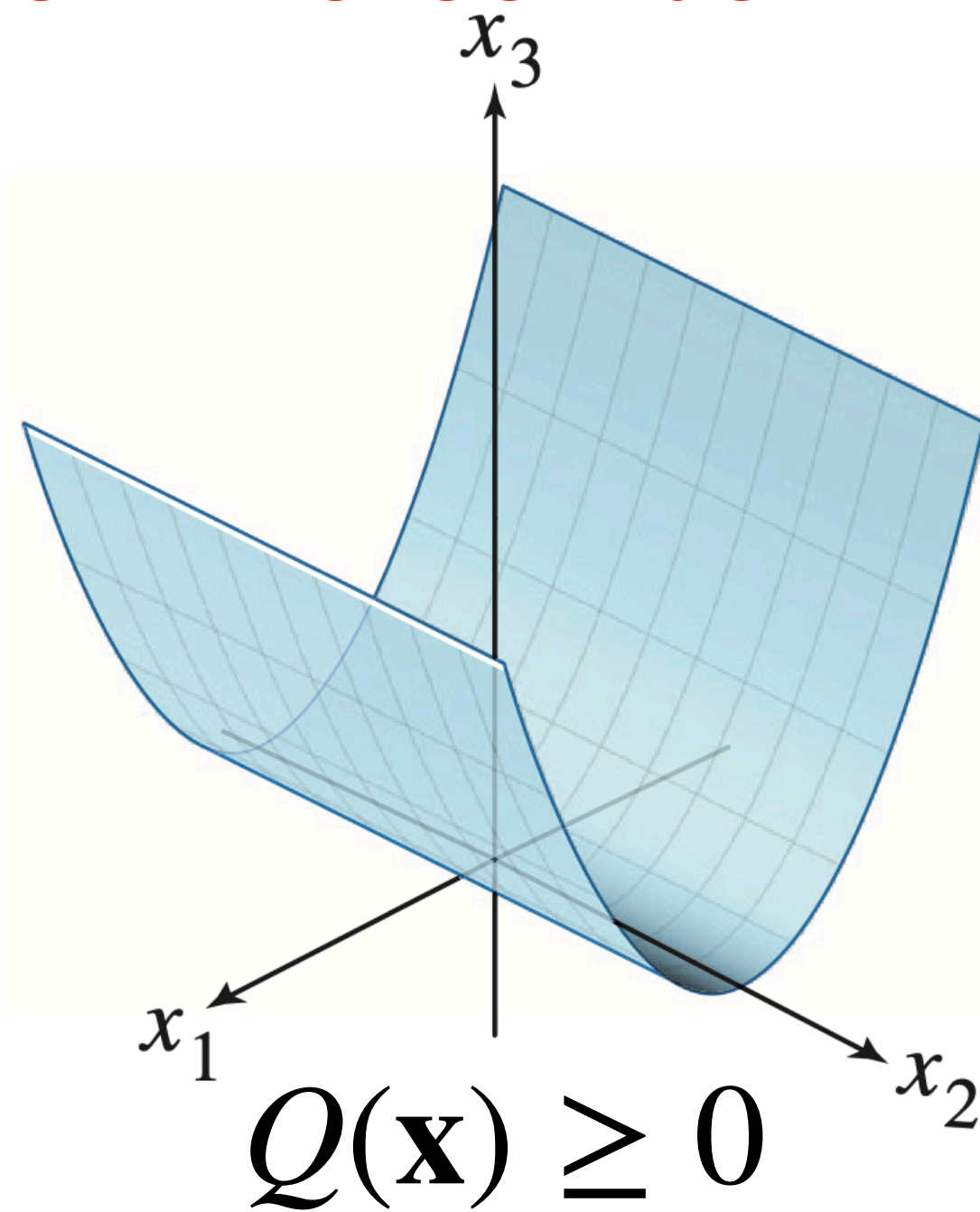
Definiteness

all nonneg. eigenvals
positive semidefinite

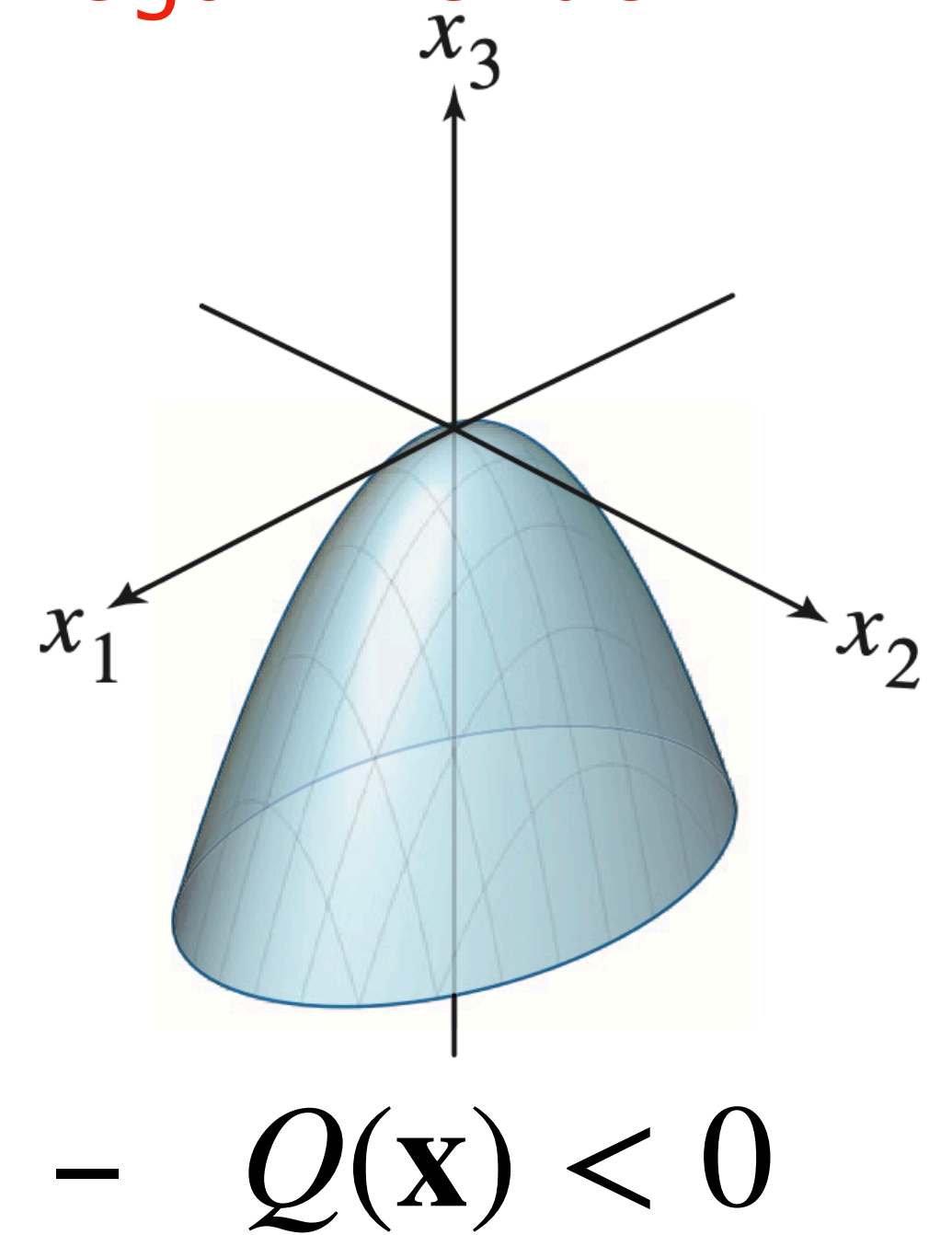
all neg. eigenvals
negative definite



positive definite
all pos. eigenvals



indefinite
pos. and neg. eigenvals



You should know how to...

- » Determine the underlying symmetric matrix of a quadratic form
- » Expand a quadratic form written as $\mathbf{x}^T A \mathbf{x}$ into a multivariate polynomial
- » Determine the definiteness of the quadratic form/symmetric matrix

Constrained Optimization

In General

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Given a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and a set of vectors X from $\mathbb{R}^n$ the **constrained minimization problem** for f over X is the problem of determining

$$\min_{\mathbf{v} \in X} f(\mathbf{v}) \quad \text{or} \quad \max_{\mathbf{v} \in X} f(\mathbf{v})$$

In General

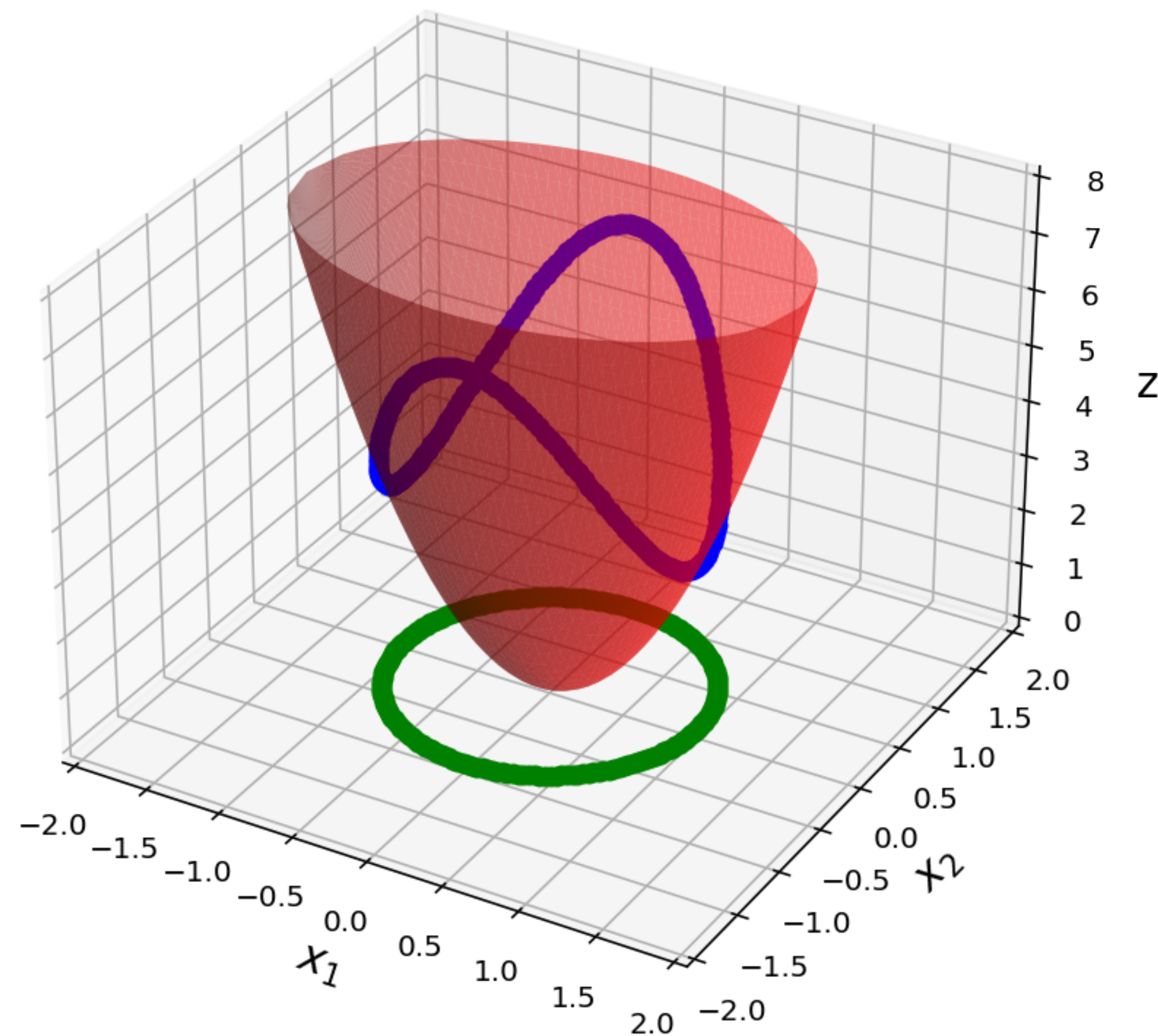
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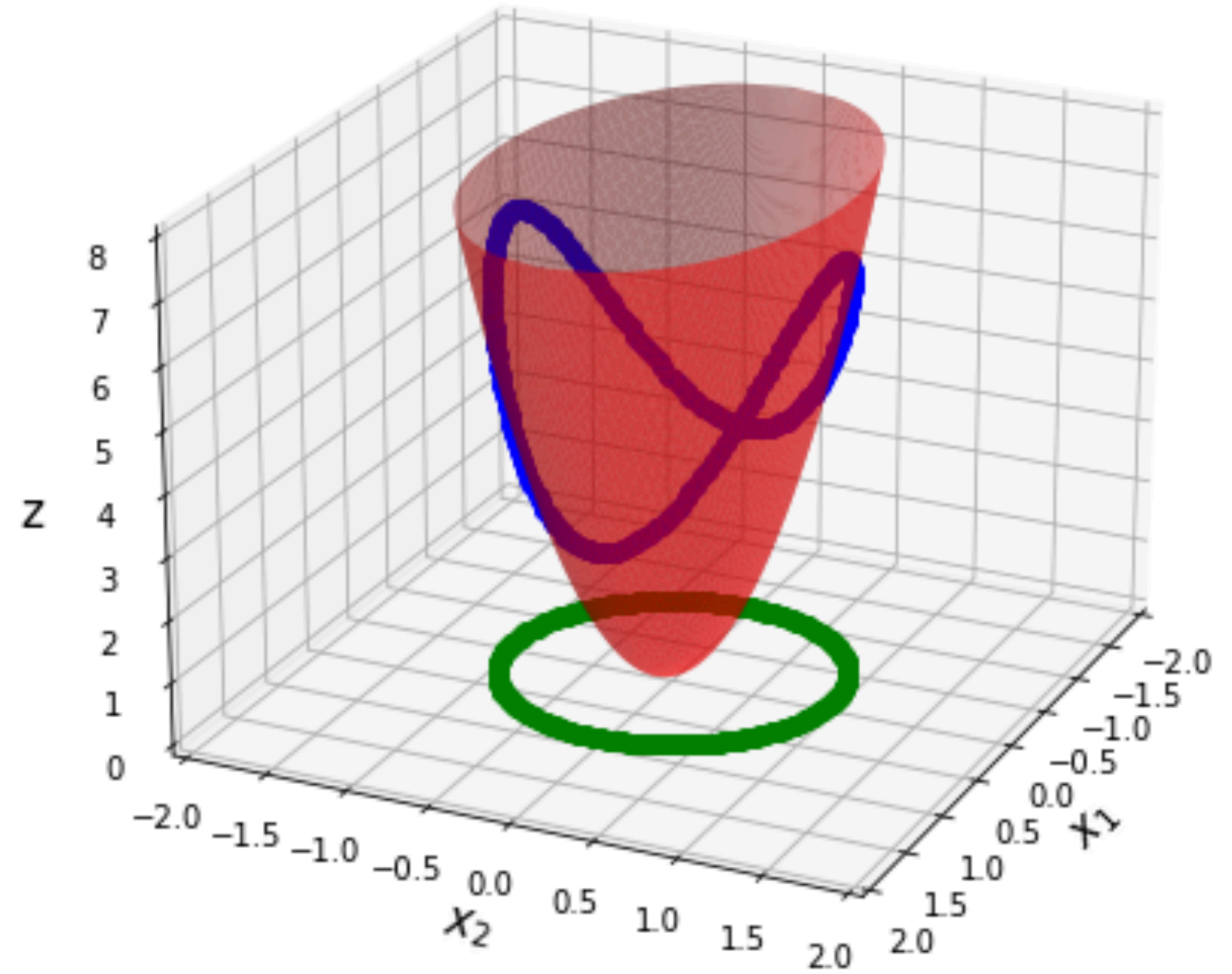
Determine the smallest value of $f(\mathbf{v})$ subject to choosing a vector in X

Constrained Optimization for Quadratic Forms and Unit Vectors

mini/maximize $\mathbf{x}^T A \mathbf{x}$ subject to $\|\mathbf{x}\| = 1$



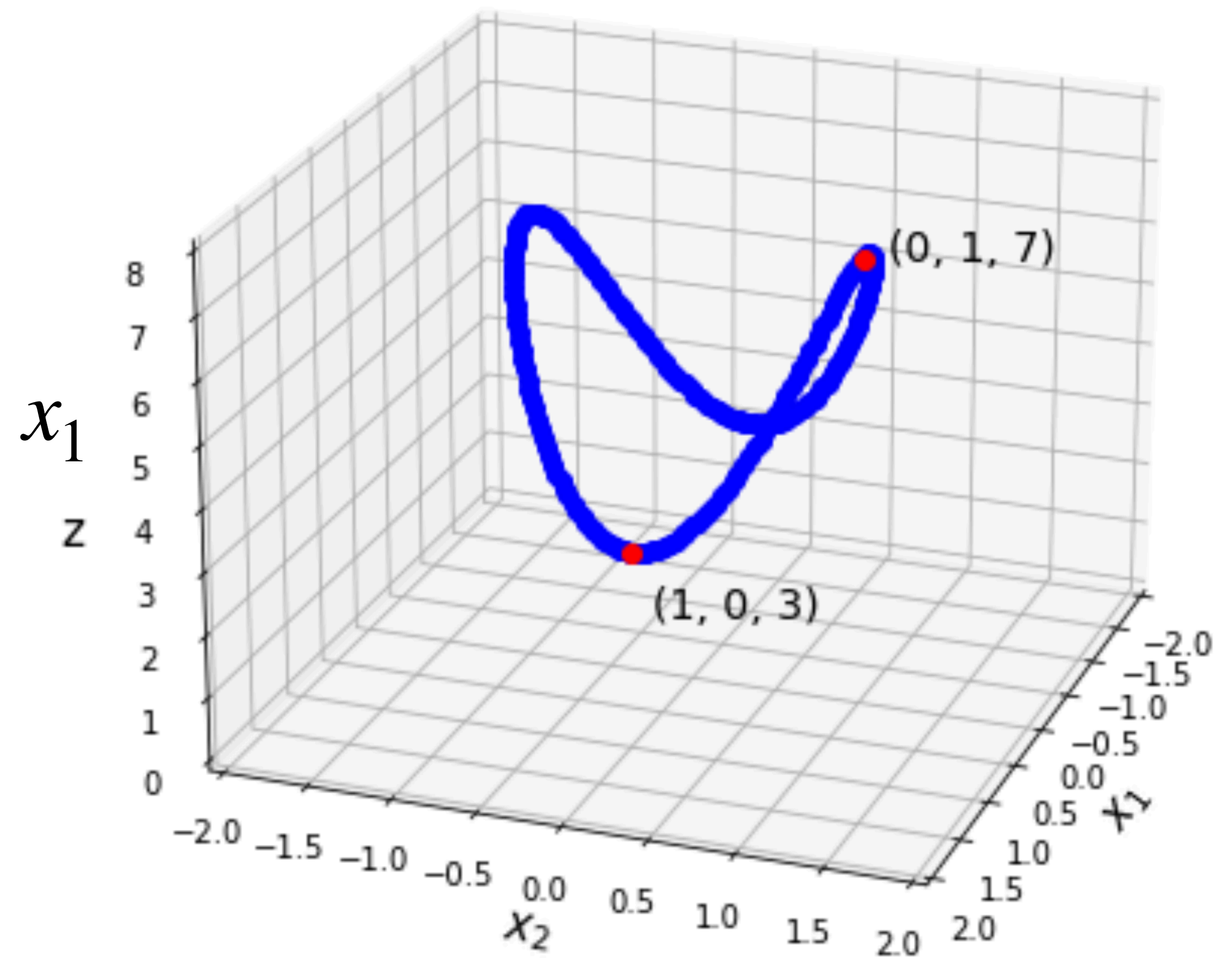
Example: $3x_1^2 + 7x_2^2$



Example: $3x_1^2 + 7x_2^2$

$$Q(x_1, x_2) = \vec{x}^T \begin{bmatrix} 3 & 0 \\ 0 & 7 \end{bmatrix} \vec{x}$$

$$\lambda_1 = 3 \quad \lambda_2 = 7$$



Constrained Optimization and Eigenvalues

Thm. For a symmetric matrix A with *largest* eigenvalue λ_1 and *smallest* eigenvalue λ_n

$$\max_{\|\mathbf{x}\|=1} \mathbf{x}^T A \mathbf{x} = \lambda_1$$

$$\min_{\|\mathbf{x}\|=1} \mathbf{x}^T A \mathbf{x} = \lambda_n$$

No matter what A looks like, this holds

You should know how to...

» Determine $\max_{\|\mathbf{x}\|=1} \mathbf{x}^T A \mathbf{x}$ and $\min_{\|\mathbf{x}\|=1} \mathbf{x}^T A \mathbf{x}$ for a given symmetric matrix A

» Determine $\max_{\|\mathbf{x}\|=1} Q(\mathbf{x})$ and $\min_{\|\mathbf{x}\|=1} Q(\mathbf{x})$ for a given quadratic form $Q(\mathbf{x})$

workshop